PARISHRAM 2.0 2026 (Class 12th)

Maths

DPP:06

Relations and Functions

Q1 Let set A has 7 elements and set B has 8 elements, then number of one-one mapping that can be defined from A to B is

(A) 56 (B) 5760

(C) 40320 (D) 192

Q2 Let N be the set of natural numbers and the function f : N $\rightarrow$ N be defined by f(n) = 2n + 3 $\forall \mathbf{n} \in \mathbf{N}$.Then f is

(A) surjective (B) injective

(C) bijective (D) none of these

Q3 Let $\mathbf{f} : \mathbf{R} \rightarrow \mathbf{R}$ be defined as $f(x) = x^4$. Choose the correct answer :

(A) f is one-one onto

(B) f is many-one onto

(C) f is one-one but not onto

(D) f is neither one-one nor onto

Q4 The function $f : R \rightarrow R : f(x) = x^3$ is

(A) one-one and onto

(B) one-one and into

(C) many-one and onto

(D) many-one and into

Q5 Let $f : R - \{8\} \rightarrow R$ be a function defined by $f(x) = \frac{x-7}{x-8}$, then

(A) f is One-One Onto

(B) f is One-One Into

(C) f is Many-One Onto

(D) f is Many-One Into

Q6 The function $f : R \rightarrow R,\ f(x) = \cos x$ is

(A) one-one and into

(B) one-one and onto

(C) many-one and into

(D) many-one and onto

Q7 Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be defined by $f(x) = \frac{x^2-8}{x^2+2}$, then f is

(A) one-one but not onto

(B) one-one and onto

(C) onto but not one-one

(D) neither one-one nor onto

Q8 The $f : N \rightarrow N,\ f(x) = x^2 + x + 1$ is

(A) one-one and onto

(B) one-one and into

(C) many-one and onto

(D) many-one and into

Q9 The function

$$f : R \rightarrow R,\ f(x) = \begin{cases} 1, & \text{if } x \text{ is rational} \\ -1, & \text{if } x \text{ is irrational} \end{cases}$$

is

(A) one-one and into

(B) one-one and onto

(C) many-one and into

(D) many-one and onto

Q10 If the set A contains 5 elements and the set B contains 6 elements, then the number of one-one and onto mappings from A to B is

(A) 720 (B) 120

(C) 0 (D) None of these

Answer Key

Q1	C
Q2	B
Q3	D
Q4	A
Q5	B
Q6	C
Q7	D
Q8	B
Q9	C
Q10	C

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

As we know when n(A) = m and n(B) = n

then no. of one -one functions $=^{n} P_{m}$

Given, n(A)=7 and n(B)=8

$\therefore$ no. of one-one mapping from A to B $=^{8} P_{7}$

$= \frac{8!}{(8-7)!}$

= 8!

= 40320

Video Solution:

Q2 Text Solution:

One-one

$f(n_1) = f(n_2)$

$\Rightarrow 2n_1 + 3 = 2n_2 + 3$

$\Rightarrow 2n_1 = 2n_2$

$\Rightarrow n_1 = n_2$

$\Rightarrow$f is one - one.

Onto

Let y = f(n) = 2n + 3

$\Rightarrow$ y – 3 =2n

$\Rightarrow n = \frac{y-3}{2} \notin N$

$\therefore$f is not onto.

Video Solution:

Q3 Text Solution:

f: $R \to R$ is defined as f(x) = x^4

One-one

f(1) = f(–1) = 1.

Here f(1) = f(–1) but –1 $\neq$1, Hence f not one-one

Onto

For $-2 \in R(\text{Co-domain})$ there does not exist pre-image in R (Domain)

$\therefore$ f is not onto.

Video Solution:

Q4 Text Solution:

One - one

let $x_1, x_2 \in R$

such that $f(x_1) = f(x_2)$

$\Rightarrow x_1^3 = x_2^3$

$\Rightarrow x_1 = x_2$

$\therefore f$ is one-one

Onto

let $f(x) = y$

$\Rightarrow x^3 = y$

$\Rightarrow x = y^{1/3}$

As for every $y \in R$ there exists

$x\left(= y^{1/3}\right) \in R$

such that $f(x) = y \therefore f$ is onto

Video Solution:

Q5 Text Solution:

One-one

let $x_1, x_2 \in R$

such that $f(x_1) = f(x_2)$

$\Rightarrow \frac{x_1-7}{x_1-8}=\frac{x_2-7}{x_2-8}$

$\Rightarrow x_1x_2-8x_1-7x_2+56=x_1x_2-7x_1-8x_2+56$

$\Rightarrow 8x_2-7x_2=8x_1-7x_1$

$\Rightarrow x_2=x_1$

$\Rightarrow x_1=x_2$

$\therefore f$ is one - one function

Onto

f is not onto i.e. into function

$\because$ for $1 \in R$ (Co-domain)

there does not exist $x \in \mathbf{R}$ (Domain)

such that $f(x)=1$

Video Solution:

Q6 **Text Solution:**

One-one

f is not one-one i.e. many one

$\because f(0)=f(2\pi)=1$

But $0 \neq 2\pi$

Onto

f is not onto function i.e. into function

$\because$Range of $f(x)$ = $[-1,1] \neq$ R (Co-domain)

Video Solution:

Q7 **Text Solution:**

One-one

Given, $f\left(x\right)=\frac{x^2-8}{x^2+2}$

$f\left(3\right)=\frac{9-8}{9+2}=\frac{1}{11}$

$f\left(-3\right)=\frac{9-8}{9+2}=\frac{1}{11}$

Here $f(3)=f(-3)$ but 3 $\neq$ -3.

$\Rightarrow f$ is not one-one.

Onto

Now, $f\left(x\right)=\frac{x^2-8}{x^2+2}=1-\frac{10}{x^2+2}$

So, minimum value of $f(x)$=1−5 = −4

$\therefore$ It is not onto.

Hence, it is neither one-one nor onto.

Video Solution:

Q8 **Text Solution:**

One-one

let $x_1, x_2 \in N$

such that $f(x_1)=f(x_2)$

$\Rightarrow x_1^2+x_1+1=x_2^2+x_2+1$

$\Rightarrow x_1^2-x_2^2+x_1-x_2=0$

$\Rightarrow (x_1-x_2)(x_1+x_2)+x_1-x_2=0$

$\Rightarrow (x_1-x_2)(x_1+x_2+1)=0$

$\Rightarrow x_1-x_2$

$=0 (\because x_1+x_2+1 \neq 0 \text{ as } x_1, x_2 \in N)$

$\Rightarrow x_1=x_2$

$\therefore$ f is one-one

Onto

f is not onto i.e. into function

$\because$ for $1 \in N$ (Co-domain) there

does not exists $x \in N$ (Domain)

such that $f(x)=1$

Video Solution:

Q9 **Text Solution:**

One-one

f is not one - one

$\because \quad f(2) = f(3) = 1$

$\text{But } 2 \neq 3$

$\therefore f$ is many-one function

Onto

f is not onto function

$\because$ there exists $0 \in \mathbf{R}$ (Co-domain)

for which there is no pre-image in R (Domain)

Video Solution:

Q10 **Text Solution:**

Total no. of element in set A = 5

Total no. of element in set B = 6

As, the no. of bijection from A to B can only possible when $n(\mathrm{A}) = n(\mathrm{B})$. But here $n(\mathrm{A}) < n(\mathrm{B})$

Hence, total no. of surjection from A to B is 0.

Video Solution:

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